

Defense sheet — Assignment set 2

Scope: pricing a European basket call $V(0) = e^{-rT} E^Q[\max(w_1 S_1(T) + w_2 S_2(T) - K, 0)]$ with $w_1 = w_2 = 0.5$, $T = 1$, $r = 0.02$, $K = 100$, by three methods: Q1 Monte Carlo, Q2 moment matching, Q3 the 2D COS method. Two market setups recur in every question: **(a)** both underlyings log-normal (GBM), $S_{10}=S_{20}=100$, $\sigma_1=0.20$, $\sigma_2=0.30$, $\rho=0.30$; **(b)** S_1 log-normal, S_2 a CIR / square-root process $dS_2 = r S_2 dt + \sigma_2 \sqrt{S_2} dW_2$, $\sigma_1=0.25$, $\sigma_2=1.5$, $\rho=0.10$ in Q1 / $\rho=0$ in Q2 and Q3.

NOTE: the assignment PDF could not be rendered in this environment (no poppler/pypdf), so the “task in my own words” below is reconstructed from the in-code comments and parameters. **TODO (Bruno):** open `Computational_Finance_Assignment_2 (2) (1).pdf` and confirm the exact wording of each sub-question (especially what convergence behaviour Q1 and Q3 ask you to demonstrate).

The task (in my own words)

Price a two-asset European **basket call** under the risk-neutral measure Q . The basket is $B_T = w_1 S_1(T) + w_2 S_2(T)$; the payoff is $\max(B_T - K, 0)$, discounted at e^{-rT} .

- **Q1 — Monte Carlo.** Simulate the basket and report the price with a standard error / 95% CI.
(a) both assets GBM, correlated, exact terminal simulation. (b) S_1 GBM, S_2 a CIR square-root process under the Q -drift $r S_2$; demonstrate (i) the weak / time-discretization convergence of the Euler scheme and (ii) the statistical $O(1/\sqrt{M})$ convergence.
- **Q2 — Moment matching.** Compute the raw moments of B_T in closed form, then approximate the basket distribution by matching 2, 3, 4 moments (log-normal, shifted log-normal, Johnson SU) and price the call from each. Benchmark against a high-precision Monte Carlo price.
- **Q3 — 2D COS.** Price the basket with the two-dimensional COS method (Ruijter & Oosterlee 2012), expanding the joint density of the two state variables on $[a_1, b_1] \times [a_2, b_2]$, exploiting $\rho=0$ so the 2D density coefficients factorise into a product of 1D coefficients. Show convergence in N .

TODO (Bruno): confirm whether the PDF asks for these specific deliverables or also for, e.g., a comparison table across the three methods.

My approach and why

- **Q1:** exact GBM simulation at maturity where possible (no time grid needed); Euler-Maruyama with full truncation for the CIR leg because exact joint simulation is unavailable when $\rho \neq 0$.
- **Q2:** derive $E[B_T^k]$ analytically from the binomial expansion of $(w_1 S_1 + w_2 S_2)^k$ and the cross moments of correlated log-normals (and the non-central chi-square moments for the CIR leg), then fit successively richer distributions to more moments.
- **Q3:** 1D characteristic functions per marginal, cumulant-rule truncation range per marginal, payoff coefficients by 2D trapezoidal quadrature, assembled as $\chi_1^T V \chi_2$.

TODO (Bruno): for each method, state in one sentence *why* it is the right tool and what each buys you (MC = unbiased baseline, moment matching = cheap closed-form approximation, COS = spectral accuracy). Do not let the examiner hear a generic answer.

Key code sections (file + what it does + why I wrote it that way)

Q1 — basket Monte Carlo (`Q1_basket_MC (1).py`)

- **`simulate_basket_lognormal(M)` — lines 17-42.** Q1(a). Draws two standard normals (Z_1, Z_c , lines 27-28) and correlates them by Cholesky: $Z_2 = \rho Z_1 + \sqrt{1-\rho^2} Z_c$ (line 29). Simulates terminal prices with the **exact GBM solution** at T (lines 32-33), forms the

basket and discounted payoff (lines 36-37), and returns the MC mean and standard error $\text{std}(\text{ddof}=1)/\text{sqrt}(M)$ (lines 40-41).

- **Convergence loop — lines 48-63.** Prices over $M = 2000 \dots 1e6$ (line 48), then estimates the convergence rate as the OLS slope of $\log(\text{SE})$ vs $\log(M)$ (line 62), expected -0.5 .
- **simulate_basket_cir(M, n_steps) — lines 93-125.** Q1(b). Time-steps with $h = T/n_steps$ (line 107). S1 exact GBM step (line 118); S2 **Euler-Maruyama** step with **full truncation** $S2p = \max(S2, 0)$ inside drift and diffusion (lines 120-121), so the square root is always defined.
- **Weak-error test — lines 134-185.** Reuses the **same fine Brownian increments** across all coarse grids (basket_on_grid, lines 144-159; increments summed into blocks at lines 151-152) to isolate the discretization bias from MC noise. Fits the weak order as the slope of $\log|\text{bias}|$ vs $\log h$ on the coarse grids (line 184), expected 1.0 .
- **Statistical-error test — lines 198-210.** Fixes the grid, varies M , fits the SE slope (line 209).

TODO (Bruno): justify (i) full truncation vs reflection / other CIR fixes; (ii) why you reuse the same increments across grids for the weak-error test; (iii) the Q-drift argument at lines 86-91 (physical CIR drift $\kappa(\theta - S2)$ replaced by $r - S2$ because the discounted price is a Q-martingale).

Q2 — moment matching (Q2_moment_matching (1).py)

- **lognormal_cross(j, m, ...) — lines 20-29.** Cross raw moment $E[S1^j S2^m]$ for two correlated log-normals, from $E[\exp(N)] = \exp(\text{mean} + 0.5 \text{ var})$ with the combined log-return mean/variance/cov at lines 26-29.
- **basket_moments_LN(...) — lines 31-37.** Raw basket moments $E[B_T^k]$ via the binomial expansion $\sum_j C(k, j) w1^j w2^{k-j} E[S1^j S2^{k-j}]$.
- **cir_S2_rawmoments(...) — lines 39-53.** Raw moments of $S2(T) = c \cdot Z$, $Z \sim \chi'^2_{2\theta}(\lambda)$ (zero-dof non-central chi-square), using cumulants $\kappa_n = 2^{n-1} n! \lambda$ (line 49) and the raw-from-cumulant recursion (lines 50-52). c and λ from the technical hint at lines 46-48.
- **basket_moments_CIR(...) — lines 55-64.** Combines the LN-S1 and CIR-S2 marginal moments under independence ($\rho=0$), so joint moments factorise (line 63).
- **central_from_raw(mom) — lines 66-75.** Converts raw moments to mean, variance, skewness, kurtosis.
- **price_2moments — lines 78-89.** Fits a log-normal matching $m1, m2$; Black's formula with $F = m1$, total variance $s^2 = \log(m2/m1^2)$.
- **price_3moments — lines 91-107.** Shifted log-normal $B = \gamma + L$; inverts the log-normal skewness $(w+2)\text{sqrt}(w-1) = g1$ for w by brentq (line 99), recovers shift and shifted strike.
- **price_4moments — lines 109-139.** Johnson SU; solves the 2×2 system for (δ, γ) from skewness and kurtosis by least_squares with Gauss-Hermite moments of $\sinh(U_moments)$, lines 120-125; residual lines 127-132; solve line 134), then prices by Gauss-Hermite quadrature (line 139). GH nodes/weights set at lines 16-17.
- **MC benchmarks — mc_benchmark_LN lines 144-159, mc_benchmark_CIR lines 161-180.** Exact simulation with antithetic variates; the CIR benchmark samples the zero-dof non-central chi-square via a Poisson mixture of Gammas (lines 171-173).

TODO (Bruno): be ready to (i) derive the cross-moment formula at line 29 by hand; (ii) derive the non-central chi-square cumulants $\kappa_n = 2^{n-1} n! \lambda$; (iii) explain *why* more matched moments need not monotonically reduce the pricing error.

Q3 — basket COS (Q3_basket_COS (1).py)

- **chi_coeffs(phi, a, b, N) — lines 22-32.** 1D density cosine coefficients $\chi_k = \text{Re}\{\phi(\omega_k) \exp(-i \omega_k a)\}$, $\omega_k = k \pi / (b-a)$ (lines 29-30), with the $k=0$ term halved (line 31) to realise the primed summation.
- **payoff_coeffs_2d(...) — lines 34-54.** Payoff cosine coefficients $V_{\{k1, k2\}}$: builds the payoff matrix $P = \max(w1 g1(y1) + w2 g2(y2) - K, 0)$ on a dense grid (line 49) and contracts it with cosine/trapezoidal-weight matrices $C1, C2$ (lines 52-53) as $C1^T P C2$ (line 54). Trapezoidal weights at lines 45-46; normalisation $(2/(b1-a1))(2/(b2-a2))$ at line 54.

- **price_2dCOS(...)** — **lines 56-63**. Assembles $e^{-rT} \chi_1^T V \chi_2$ (line 63).
- **cumulant_range(k1, k2, k4, L=12)** — **lines 65-71**. **The truncation range**. Returns $[k1 - \text{width}, k1 + \text{width}]$ with $\text{width} = L \cdot \sqrt{|k2| + |k4|}$ — **lines 70-71**. This is the COS-paper cumulant rule; $L=12$ is the default.
- **Q3(a) characteristic functions** — **lines 79-80**. Normal log-price CFs $\phi(\omega) = \exp(i \omega \mu - 0.5 \sigma^2 T \omega^2)$ for each marginal; means at lines 76-77.
- **Q3(a) ranges** — **lines 82-83**. **cumulant_range**($\mu, \sigma^2 T, 0$) per marginal (a normal has $k1 = \text{mean}, k2 = \text{variance}, k4 = 0$).
- **Q3(b) S2 characteristic function** — **phi2_b, lines 117-119**. CF of $S_2(T) = c \cdot Z, Z \sim \chi'^2_{0}(\lambda)$: $\exp(\lambda i z / (1 - 2i z))$ with $z = \omega \cdot c$ (lines 118-119); c_b, λ_b at lines 114-116.
- **Q3(b) cumulants of S2** — **lines 122-124**. $k1 = c \cdot \lambda, k2 = c^2 \cdot 4 \cdot \lambda, k4 = c^4 \cdot 192 \cdot \lambda$ (from $\kappa_n = 2^{(n-1)} n! \lambda$ scaled by c^n).
- **Q3(b) ranges** — **lines 125-127**. $S1$ range **cumulant_range**($\mu_1_b, \sigma_1^2 T, 0$) (line 125); $S2$ range **cumulant_range**($k1_{S2}, k2_{S2}, k4_{S2}, L=12$) (line 126), then **clamped to be non-negative** $a2_b = \max(a2_b, 0.0)$ (line 127) because $S2 \geq 0$.
- **Payoff maps** — **lines 84-85, 128-129**. $g = \exp$ for log-price state variables; $g2_b = \text{identity}$ in Q3(b) because the $S2$ state variable is the *level*, not the log.
- **Convergence loops** — **lines 88-104 (a), 131-145 (b)**. Compare COS prices over $N = 8 \dots 64$ against a converged $N = 256$ reference.

Design decisions I must justify

- **Cholesky correlation** ($Z_2 = \rho Z_1 + \sqrt{1-\rho^2} Z_c$) — Q1 lines 29, 116; Q2 line 152. **TODO (Bruno)**: why this is exactly the 2×2 Cholesky factor.
- **Full truncation** of the CIR scheme — Q1 lines 120-121. **TODO (Bruno)**: why truncation (not reflection) and what bias it introduces.
- **Reusing fine Brownian increments across grids** for the weak-error test — Q1 lines 144-169. **TODO (Bruno)**: why this cleanly isolates discretization bias from MC noise.
- **Antithetic variates** in the Q2 benchmarks — Q2 lines 154-158, 176-179. **TODO (Bruno)**: why antithetics reduce variance here and when they fail.
- **Choice of approximating families** (LN / shifted-LN / Johnson SU) — Q2 lines 78-139. **TODO (Bruno)**: why each family is the natural one for 2 / 3 / 4 matched moments.
- **$\rho = 0$ factorisation** of the 2D density coefficients — Q3 lines 13-20, 60-63. **TODO (Bruno)**: show why independence makes $c_{\{k1,k2\}} = \chi_{\{k1\}} \chi_{\{k2\}}$ and what changes if $\rho \neq 0$.
- **$L = 12$ in the cumulant rule** — Q3 line 65, used at lines 82-83, 125-126. **TODO (Bruno)**: justify $L = 12$ (the paper often uses 8-10); relate to the bug below.
- **Trapezoidal quadrature for $V_{\{k1,k2\}}$** instead of a closed form — Q3 lines 34-54. **TODO (Bruno)**: why no analytic V_k here (basket payoff couples the two state variables), and how Ngrid was chosen.

Results and what they mean

TODO (Bruno): run the three scripts and fill in the actual numbers: - Q1(a) price + 95% CI, and the fitted SE slope (should be ≈ -0.5 , line 63). - Q1(b) price + CI, the fitted weak order (≈ 1.0 , line 184), the fitted SE slope (line 209). - Q2(a) and Q2(b): basket mean/std/skew/kurtosis, MC benchmark, and the 2/3/4-moment prices with absolute errors (printed at lines 196-198, 214-216) — explain why error generally falls with more moments and any exception. - Q3(a)/(b): converged COS price and the error/ratio table (lines 98-104, 140-145) — explain whether convergence looks spectral (exponential) and how the ratios behave.

The hardest question Fang could ask here — and my answer

Most likely deep-dive (highest probability): the COS truncation range $[a,b]$. Fang co-invented COS; the cumulant truncation rule is her result, and this assignment had a real bug there.

She will go straight to `Q3_basket_COS (1).py:65-71` (cumulant_range) and its call sites :82-83 (Q3a) and :125-127 (Q3b).

What the code factually does: `cumulant_range` returns $[k1 - L \cdot \sqrt{|k2| + \sqrt{|k4|}}], k1 + L \cdot \sqrt{|k2| + \sqrt{|k4|}}]$ (`Q3_basket_COS (1).py:70-71`), with $L = 12$. For a normal marginal $k4 = 0$, so the half-width is $L \cdot \sqrt{\text{variance}}$ (:82-83). For the CIR leg, the half-width uses the non-central chi-square cumulants $k2 = 4 \cdot c^2 \cdot \text{lambda}$, $k4 = 192 \cdot c^4 \cdot \text{lambda}$ (:123-124), and the lower end is clamped to 0 (:127).

The bug (special focus): a too-small c2 made [a,b] too narrow at short maturities. The half-width scales like $\sqrt{k2}$, and $k2$ is the variance term — for a normal marginal $k2 = \sigma^2 T$ (:82-83, :125), which **shrinks with T**. At short maturity $c2$ is small, so $\text{width} = L \cdot \sqrt{c2 + \sqrt{c4}}$ (`Q3_basket_COS (1).py:70`) collapses and $[a,b]$ becomes too narrow to contain the support of the density / payoff, truncating mass and corrupting the price. The fix lives at the range construction — the `cumulant_range` formula at :70 and the L value at :65 (and the non-negativity clamp at :127).

TODO (Bruno): the reasoning of the fix — fill this in yourself. Specifically: 1. What symptom did you see (wrong/unstable price, or error not converging) and at which T ? 2. What exactly did you change — raise L , add the $\sqrt{k4}$ term, widen the floor, or change how $c2$ is fed in for short T ? Point to the precise line you edited (:65 $L=12$, :70 the width formula, or :127 the clamp). 3. Why $L \cdot \sqrt{c2 + \sqrt{c4}}$ (and not just $L \cdot \sqrt{c2}$) is the right safeguard when $c2$ is tiny but $c4 > 0$, and why a fixed wide interval is *not* a free lunch (more terms N needed for the same accuracy, since COS error depends on both truncation and series length).

This is the single question most worth over-preparing.

“Show me in your code where you do X” — anticipated spots

- “...where you compute the truncation range $[a,b]$.” `Q3_basket_COS (1).py:65-71` (`cumulant_range`), called at :82-83 (Q3a) and :125-127 (Q3b, with the $\max(a_2b, 0)$ clamp at :127).
- “...where a small $c2$ makes $[a,b]$ too narrow, and how you fixed it.” `Q3_basket_COS (1).py:70` ($\text{width} = L \cdot \sqrt{\text{abs}(k2) + \sqrt{\text{abs}(k4)}}$) and :65 ($L=12$). **TODO (Bruno):** name the line you edited.
- “...where you define the characteristic function.” Q3(a) :79-80; Q3(b) CIR leg `phi2_b` :117-119.
- “...where you build the cosine density coefficients χ_k .” `chi_coeffs` `Q3_basket_COS (1).py:22-32` (note the $\chi[0] *= 0.5$ primed-sum term at :31).
- “...where you build the payoff coefficients $V_{\{k1,k2\}}$.” `payoff_coeffs_2d` :34-54 (assembly :54).
- “...where the 2D price is assembled.” `price_2dCOS` :56-63 ($\chi1 @ V @ \chi2$ at :63).
- “...where the cumulants of the CIR leg come from.” `Q3_basket_COS (1).py:122-124`; same moments in `Q2_moment_matching (1).py:39-53`.
- “...where you correlate the Brownian motions.” `Q1_basket_MC (1).py:29` and :116.
- “...where you do the Euler step for $S2$.” `Q1_basket_MC (1).py:120-121` (full truncation).
- “...where you isolate the discretization bias.” `Q1_basket_MC (1).py:144-169`.
- “...where you match 2 / 3 / 4 moments.” `price_2moments` :78-89, `price_3moments` :91-107, `price_4moments` :109-139 in `Q2_moment_matching (1).py`.
- “...where the MC benchmark uses antithetics.” `Q2_moment_matching (1).py:154-158` and :176-179.